TOPFAITH UNIVERSITY
MKPATAK, AKWA IBOM STATE, NIGERIA

B.Sc. DEGREE EXAMINATION

FACULTY: COMPUTING & APPLIED SCIENCES
DEPARTMENT: MATHEMATICS
SESSION: 2023/2024 **SEMESTER**: PRIMO
COURSE CODE: MTH 101 **CREDIT UNIT**: 2
COURSE TITLE: ELEMENTARY MATHEMATICS I **TIME**: 2 HOURS
INSTRUCTION: ANSWER ANY THREE (3) QUESTIONS

---

1. a) There are 120 students who took either Geography or Economics in an examination. 65 took Geography and 57 took Economics. Only 17 students took Geography but not Economics. With a well-illustrated Venn diagram, calculate how many students took:
   i) Economics but not Geography
   ii) neither Geography nor Economics
   iii) both subjects (18marks)

   b) Factorize $8x^2 + 2x - 3 = 0$ (5.5marks)

   [23.5 MARKS]

2. a) State the axioms of mathematical induction. (5.5marks)

   b) Prove that the statement $1 + 3 + 5 + \ldots + (2n-1) = n^2$ holds for all positive integers *n*. (18marks)

   [23.5 MARKS]

3. a) Insert three arithmetic means between 19 and 35. (11.5marks)

   b) Given the complex numbers $z_1 = -\sqrt{3} + 33i$ and $z_2 = -2 + 2i$, find:

   $i)\ z_1 z_2$ $ii)\ \frac{z_1}{z_2}$ (12marks)

   [23.5 MARKS]

4. a) Find the first five terms of the expansion $(1 - x)^{10}$ in ascending powers of x. (15marks)

   b) Using the expansion in (a) above, find the value of $(0.98)^{10}$ correct to 5places of decimals. (8.5marks)

   [23.5 MARKS]

5. a) Evaluate the following: $i)\ (\sqrt{3} + i)^6$ $ii)\ (-1 - i)^7$ (14.5marks)

   b) Change the following angles in radians to degrees:

   i) 2.5^c ii) 0.8^c iii) $\frac{1}{3}\pi^c$ (9marks)

   [23.5 MARKS]

TOPFAITH UNIVERSITY
MKPATAK, AKWA IBOM STATE, NIGERIA

B.Sc. DEGREE EXAMINATION

FACULTY: COMPUTING & APPLIED SCIENCES
DEPARTMENT: MATHEMATICS **SEMESTER**: PRIMO
SESSION: 2023/2024 **COURSE CODE**: MTH 101
CREDIT UNIT: 2
COURSE TITLE: ELEMENTARY MATHEMATICS I **TIME**: 2 HOURS
INSTRUCTION: ANSWER ANY THREE (3) QUESTIONS

MARKING SCHEME

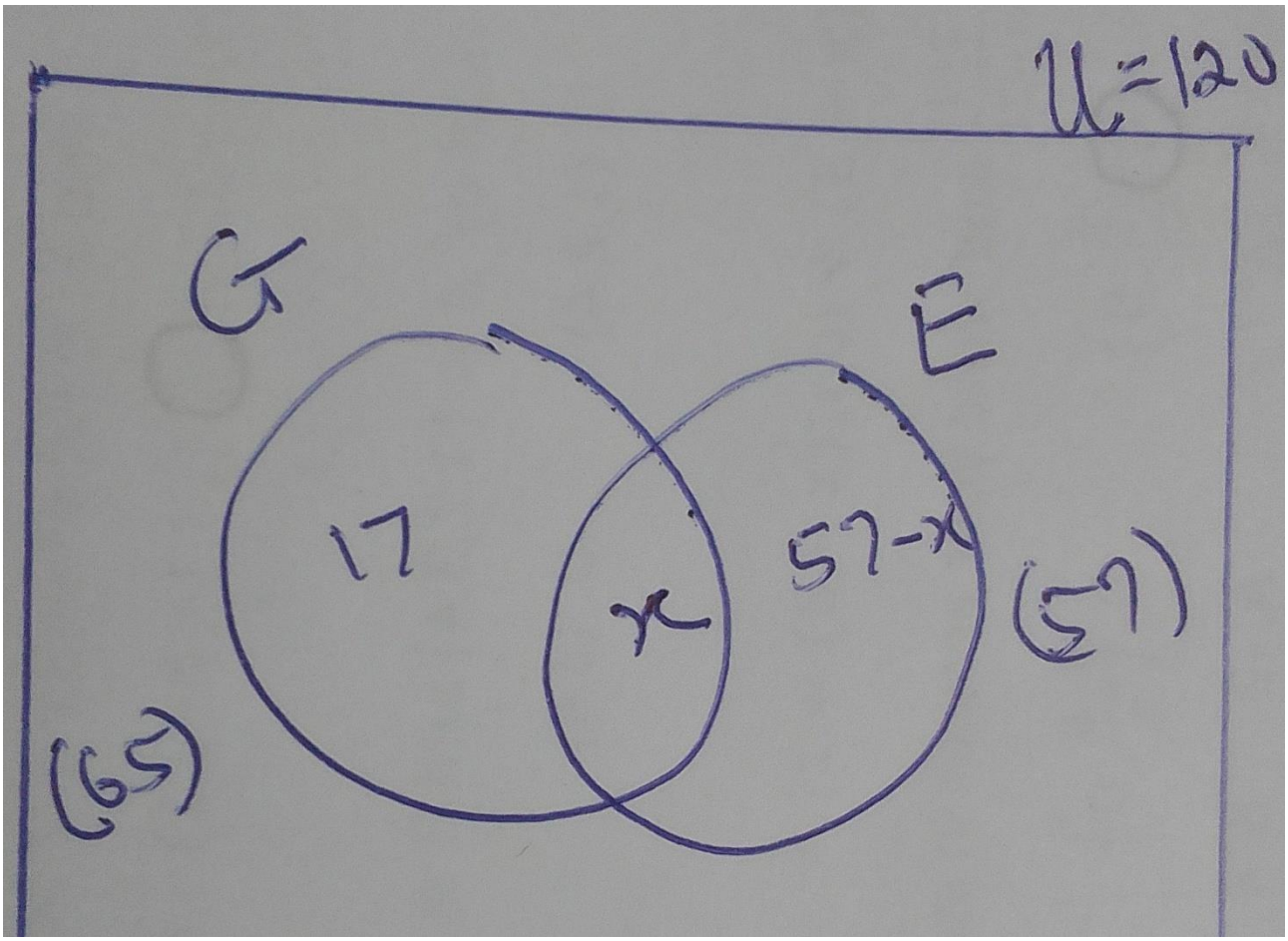
Question No. 1

a) Let U represents the total number of studnts in the school

Let G represents the number of students who took Geography

Let E represents the number of students who took Economics **3mks**

$n(U)=120,\ n(E)=57,\ n(G)=65, n(GnE')=17,\ n(GnE)=x$ **1.5mks**



4.5mks

iii) $n(GnE) = x$

$x = 65 - 17$

$\therefore x = 48$ **3mks**

i) $n(EnG') = n(E\ only)$

$n(E\ only) = 57 - x = 57 - 48$

$\therefore n(E\ only) = 9$ **3mks**

ii) $n(Neither\ Geography\ nor\ Economics) = n(u) - n(G \cup E)$

$= 120 - (17 + 57)$

$= 46$ **3mks**

b. Factorizing $8x^2 + 2x - 3 = 0$

$8x^2 + 6x - 2x - 3 = 0$

2x (4x +3) -1(4x +3) = 0 **2.5mks**

(4x+3)(2x -1) = 0 **1mks**

Hence, x = ½ or -3/4 **2mks**

[Total = 23.5marks]

Question No. 2

a. **Axioms of Mathematical Induction**

Let S be a set of positive integers. If

i 1Є S and **2mks**

ii K Є S implies that k+1 Є S, **2mks**

Then all the positive integers are in S. **1.5mks**

b. $1 + 3 + 5 + \ldots + (2n-1) = n^2$ holds for all positive integers n

when n =1, then

$2(1) - 1 = 1^2$ **1.5mks**

2-1=1 implying that 1=1 which is true

When n=2,

$2(2) - 1 = 2^2$ **1.5mks**

4 - 1 = 4 and by the axiom of induction n=2 is true **2mks**

When n=k

$2(k-1) = k^2$ ---------- (1) **2mks**

When n=k+1

$2[(k+1) -1] = (k+1)^2$ ---------- (2) **2mks**

Combining equations (1) and (2)

$K^2 + 2[(k+1)-1] = (k+1)^2$ **2mks**

$K^2 + 2k+1 = (k+1)(k+1)$ **2mks**

$K^2 + 2k+1 = K^2 + 2k+1$ which is true. **2mks**

Hence, by the axiom of induction, all positive integers are in S. **3mks**

[Total = 23.5marks]

Question No. 3

a. Let the arithmetic means be b, c and d. Then we have 19, b, c, d and 35 **1mk**

where a = 19

Recall that, $T_n = a + (n-1)d$ **1mk**

$T_5 = 19 + 4d$ **1mk**

But, $T_5 = 35$ **1mk**

Hence,

$35 = 19 + 4d$ **1mk**

$d = 16/4 = 4$ **1mk**

It follows that,

$b = a + d = 19 + 4 = 23$ **1mk**

$c = a + 2d = 19 + 2(4) = 27$ **1mk**

Also, $d = a + 3d$

$d = 19 + 3(4) = 31$. **1mk**

Thus, the three Arithmetic means are 23, 27 and 31. **2mks** (11.5marks)

b. i) $z_1 = -\sqrt{3} + 3i$, $z_2 = -2 + 2i$

$z_1z_2 = (-\sqrt{3} + 3i)(-2 + 2i)$ **2mks**

$= 2\sqrt{3} - 6 - i2\sqrt{3} - i2(3)$ **2mks**

$= 2\sqrt{3} - 6 - i(6 + 2\sqrt{3})$ **1mk**

$= -2.54 - 9.46i$ **1mk**

ii) $\frac{z_1}{z_2} = \frac{-\sqrt{3} + 3i}{-2 + 2i}$ **1mk**

rationalizing,

$\frac{z_1}{z_2} = \frac{(-\sqrt{3} + 3i)(-2 - 2i)}{(-2 + 2i)(-2 - 2i)}$ **2mks**

$\frac{z_1}{z_2} = \frac{6 + 2\sqrt{3} + i(2\sqrt{3} - 6)}{8}$ **1mk**

$$\frac{z_1}{z_2} = \frac{6+2\sqrt{3}}{8} + \frac{i(2\sqrt{3}-6)}{8}$$ **1mk**

$$\therefore \frac{z_1}{z_2} = 1.183 - 0.317i$$ **1mk**

[Total = 23.5marks]

Question No. 4

a. This is the expansion of $[1+(-x)]^{10}$, n=10

The first five terms are

$= (1)^{10} + {}^{10}C_1(1)^9(-x) + {}^{10}C_2(1)^8(-x)^2 + {}^{10}C_3(1)^7(-x)^3 + {}^{10}C_4(1)^6(-x)^4.$ **5mks**

$= 1 - 10x + \frac{10\times9}{2}x^2 - \frac{10\times9\times8}{3\times2\times1}x^3 + \frac{10\times9\times8\times7}{4\times3\times2\times1}x^4.$ **5mks**

$= 1 - 10x + 45x^2 - 120x^3 + 210x^4$ **5mks**

b. Using the above expansion, we proceed to obtain the value of $(0.98)^{10}$ correct to 5 decimal places as follows:

We will write 0.98 as 1 – 0.02 and substitute x = $0.02 = (2\times10^{-2})$ in the expansion. **2mks**

Then,

$(1-0.02)^{10} = 1 - 10(0.02) + 45(0.02)^2 - 12(0.02)^3 + 210(0.02)^4$ **2mks**

$(1-0.02)^{10} = 1 - 0.2 + 0.018 - 0.00096 + 0.0000336$ **2mks**

$(1-0.02)^{10} = 0.81707\,(to\ 5d.p)$ **2.5mks**

[Total = 23.5marks]

Question No. 5

a. i) $(\sqrt{3}+i)^6$

For $\sqrt{3}+i$, r=2, $\theta = \frac{\Pi}{6}$ **1mk**

$\sqrt{3}+i = 2(\cos\frac{\Pi}{6} + i\sin\frac{\Pi}{6})$ **1mk**

By the D'Moivre's theorem

$z^n = r^n(\cos n\theta + \sin n\theta)$ **1mk**

$(\sqrt{3}+i)^6 = 2^6(\cos 6\frac{\Pi}{6} + i\sin 6\frac{\Pi}{6})$ **1mk**

$(\sqrt{3}+i)^6 = 2^6(\cos\prod + i\sin\prod)$ **2mks**

$\therefore (\sqrt{3}+i)^6$ = -64 **1.5mks**

ii) For $-1-i,\ r=\sqrt{2}\ ,\ \theta=\frac{\prod}{4}$ **1mk**

$\therefore -1-i=\sqrt{2}(\cos\frac{\prod}{4}+i\sin\frac{\prod}{4})$ **2mks**

$(-1-i)^7=(\sqrt{2})^7(\cos 7\frac{\prod}{4}+i\sin 7\frac{\prod}{4})$ **2mks**

$=8\sqrt{2}(\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i)$ **1mk**

$=8-8i$ **1mk**

b. i) $2.5^c = 2.5\times\frac{180}{\prod}$ **2mks**

$=143.2^0$ **1mk**

ii) $0.8^c = 0.8\times\frac{180}{\prod}$ **2mks**

$= 45.8^0$ **1mk**

iii) $\frac{1}{3}\prod^c = \frac{1}{3}\prod\times\frac{180}{\prod}$ **2mks**

$= 60^0$ **1mk**

[Total = 23.5marks]